

Utkarsh Ranjan

Linkedin: [utkarsh-ranjan-3b49091b9](#)

GitHub: [github.com/utkarshrjn/](#)

Email: utkarsh.ranjan@gmail.com

Mobile: +91-993-9520-999

EDUCATION

Indian Institute of Technology Bombay, India

Bachelor of Technology (Honors) in Computer Science and Engineering

Nov 2020 - May 2024

GPA: 9.05/10.0

WORK EXPERIENCE

Stripe: Software Engineer, Product

Full-Time Software Engineering

June 2024 - Present

Stripe, Bangalore, India

Technologies: Apache Airflow, Apache Spark, React.js, Ruby, Hadoop, Trino

- Working in the **Financial Reporting** Team serving **2M** merchants, including **150K** Monthly Active Users.
- Spearheading **Agentic Contact Support** for financial reporting, aiming to integrate the Reporting tools into Stripe Agentic Assistant, projected to resolve **2,600** monthly support tickets and save **\$1.2M** annually.
- Contributed to **Microsoft's** EPA reporting system leveraging **PySpark** and **Airflow** for distributed data processing, integrating transaction data across multiple payment flows, which helped secure the **\$100M contract** over 5-years.
- Implemented **Full-stack** enhancements to All Fees report, providing fee transparency across top users (**13% PIV**).
- Reduced download dialog **p90** load time by **87%** (3s → 400ms); Enhanced error handling for **~90** dashboard pages.
- Optimized synthetic E2E tests, achieving **9×** faster performance and uncovering scalability gaps in infrastructure.

Prodigal (YC S'18): Machine Learning Intern

Machine Learning Intern | DSML Team

Summer 2023

Prodigal, Mumbai, India

- Automated Right Party Check by clustering call transcript segments into affirmation groups for reliable verification.
- Used **TF-IDF** embeddings with a semi-supervised **weighted K-Means** pipeline for accurate clustering.

RESEARCH & KEY PROJECTS

BharatGen: English Language Model

Guide: Prof. Ganesh Ramakrishnan | B.Tech Project

Spring 2024

IIT Bombay

- Supported training & evaluation of BharatGen, a **7B+ parameter** LLM covering **22 Indian languages**.
- Implemented **distributed training** using Megatron-DeepSpeed with data, tensor, and pipeline parallelism.
- Benchmarked model inference on **NVIDIA A100** and **Intel Gaudi2** accelerators across **30+** LM evaluation tasks.

RDF to Natural Language Conversion

Prof. Pushpak Bhattacharyya | R&D Project

Spring 2023

LG Soft India - IIT Bombay

- Finetuned **T5-base** on **A100 (80GB) GPU** for converting knowledge graphs in RDF (triples) format into text.
- Achieved **SOTA** performance of **58.9% BLEU**, **0.69 BLEURT** and **0.43 METEOR** scores on WebNLG'17 corpus.

DeepSynth: ML framework for Grammar-Based Program Synthesis

Guide: Nathanaël Fijalkow, CNRS | Research Internship

Summer 2022

LaBRI (INRIA), Bordeaux, France

- Enhanced the algorithm used for enumerating programs generated using probabilistic context-free grammar (**PCFG**).
- Introduced discrete probability tuples, reducing multiplication computations and achieving **93.1%** accuracy.

Static Analysis for Optimized Garbage Collection

Guide: Prof. Swaprava Nath | Course Project: Code Optimization for OO Language

Spring 2024

IIT Bombay

- Performed inter-procedural escape analysis on **Java code** to detect object lifetimes for **early garbage collection**.
- Transformed Java bytecode using the analysis, achieving **2×** faster execution on **OpenJ9** JVM interpreter-only mode.

C like Compiler

Guide: Prof. Amitabha Sanyal | Course Project: Compilers

Spring 2023

IIT Bombay

- Coded a compiler in C language from scratch using **Flex**, **Bison** and data structures like **Abstract Syntax Tree**
- Generated **Assembly** (32-bit x86 code) for C programs with suitable **register allocation** and **code optimisations**

SCHOLASTIC ACHIEVEMENT

- Achieved **All India Rank 356** in JEE-Advanced out of 240 thousand eligible candidates 2020
- Awarded **Gold Medal** for ranking among the **Top 46** Nationwide in the Indian National Chemistry Olympiad. 2020
- Achieved **All India Rank 174** and was awarded KVPY fellowship by IISc Bangalore. 2019
- Selected among the **2000 Top Global CS** students for the **Y Combinator's** invite-only AI Startup School. 2025

POSITIONS OF RESPONSIBILITY

Institute Student Mentor, Student Mentor Programme, IITB

May 2023 - June 2024

Department Academic Mentor, CSE Dept., IIT Bombay

May 2022 - July 2023

Team Member, IITB Racing Driverless Team

Nov 2020 - April 2021